Improvements to Existing Info Structure and Format

(Adapted from IMT 542 Assignment G7)

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Project Overview

My project is a python-based tool that compiles metadata from fanwork URLs from the website Archive of Our Own (Ao3) and then creates visualizations to analyze fanwork metadata. The tool accomplishes this in four steps:

1. The tool takes in a fandom webpage (example: https://archiveofourown.org/tags/Candela%20Obscura%20(Critical%20Role%20Web%20Series)/works) and scrapes the page for a list of "pagination links". The combined pagination links contain links to all the fanworks for the given Ao3 fandom.
2. For each pagination link, the tool scrapes the page for a list of fanworks.
3. For each fanwork link, the tool scrapes the page for the desired metadata, then organizes the compiled metadata into a json structure.
4. The tool takes the compiled json structure and uses pandas and plotly.express libraries to create visualizations for analysis.

All web scraping tasks use the BeautifulSoup library.

Maintaining & Measuring Quality

How will your structure promote maintaining and measuring quality easier?

Consistency - The tool scrapes metadata from html pages and compiles the metadata into a standard / commonly used information structure, specifically a JSON structure.

Completeness

- The tool's process—specifically Step 1, which generates a list of pagination links—is designed to ensure that the user can get as complete a list of fanwork links as possible.
- The functions that scrape and compile fanwork metadata use if/else statements to ensure that if a metadata field is not included by the author, the metadata field will still be populated with a null or default value in the JSON structure. For example:
    - If the author has not included any additional freeform tags, the get_additional_tags function will populate the 'Additional Tags' field with the string 'No additional tags'.
    - If there are no kudos, comments, bookmarks, or hits on a particular fanwork, the functions to retrieve these numbers will specify int(0) for the Kudos, Comments, Bookmarks, and Hits metadata fields.

Identifying Information Security Issues

How will your structure enable any information security issues to be identified and managed down the road?

Categories from "The 12 Elements of an Information Security Policy" https://www.exabeam.com/explainers/information-security/the-12-elements-of-an-information-security-policy/)

Protects sensitive data - Despite that fanworks authors have posted their works publicly / Ao3 works are generally publicly available, fanworks authors haven't directly consented to have their works scraped for the purpose of this tool / visualizing metadata. In particular, fanworks authors who have archive-locked their works (i.e. made their works accessible only to other Ao3 account holders) have not consented to have their works scraped for this tool's purpose. To address this, the tool explicitly does not scrape the unique identifier / URL for each fanwork, so those identifiers are not available in the JSON structure. The tool does generate a primary key for each fanwork's JSON entry so that each fanwork has a unique identifier, just not one tied to the original fanwork link, better preserving individual authors' privacy.

Provides a clear security statement to third parties - One of the most significant security risks for this tool involves misuse of the information / insights gleaned from the visualizations. As with all art, fanworks creators may choose to depict or explore upsetting or offensive content in their stories, and the presence of upsetting or offensive content should not be taken as a statement about the character of the fanwork authors whose works are represented in the resulting visualizations. I am including in the Github ReadMe a security statement that clearly outlines the limitations and intended use of the tool.